Actor Network Theory
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Technology in the Biosphere
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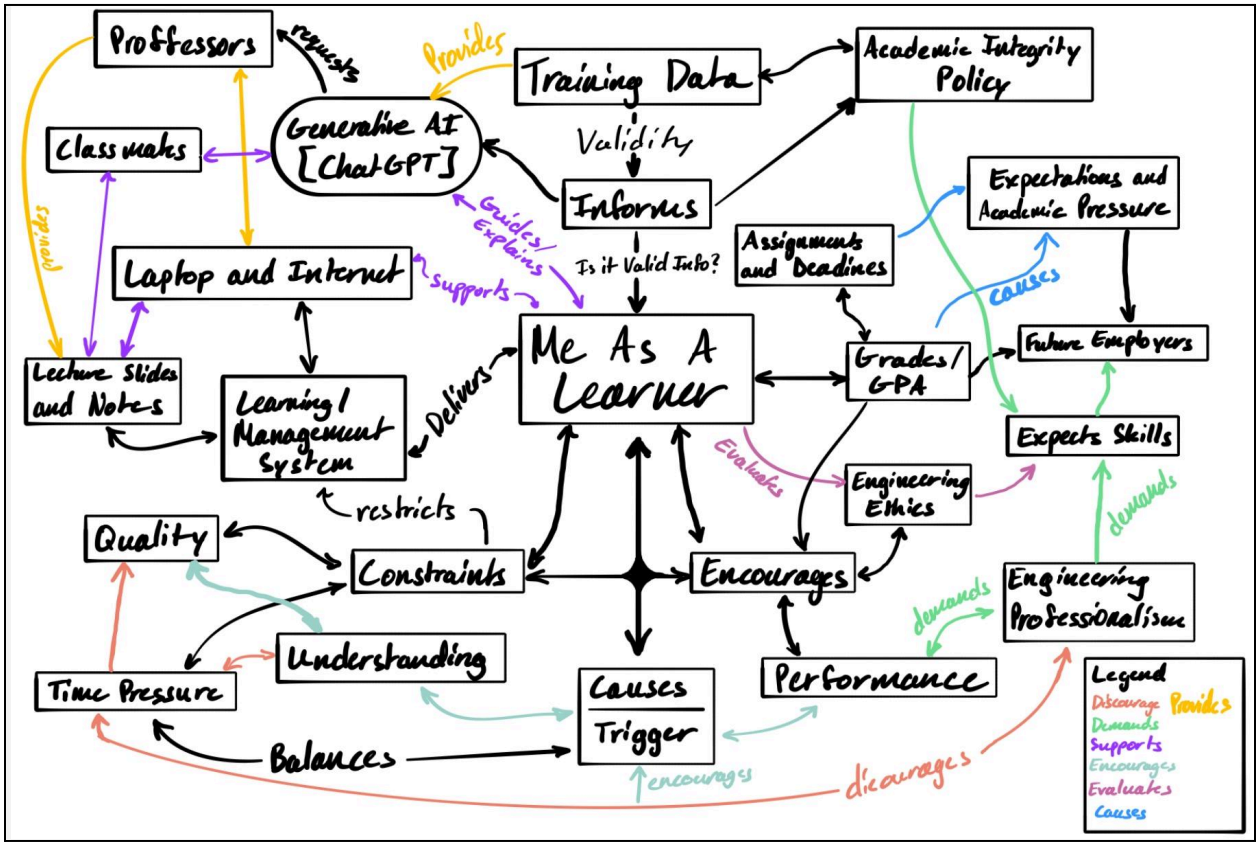
Actor Network Visual

Topic Statement: How does Generative AI impact my learning processes?



(798 words)

1. (Stable Ordering) AI-Assisted Coursework as a Durable System

In my actor-network, a stable order is the routinization of Generative AI in academic habits of schooling and learning practices. AI Mediates itself through successive translations of human and non-human actors like myself, assignments, grades, time pressure and institutional learning platforms. A pattern that has stabilized how learning is coordinated and carried out. In each case, actors are slotted into largely stable roles. Professors impose homework and make deadlines, grades act as incentives for compliance, and Generative AI produces explanations and problem-solving scaffolding. AI is more frequently used to cut down on time-consuming tasks. It can recreate fully organized work in seconds, with quality levels that often match what you would do without human alteration or even beat the original quality. As part of this setup, I am a human actor whose behaviour utilizes these tools and is both balanced and determined by the tension between understanding concepts and achieving performance.

Punctualization stabilizes the ordering of this network. The common practice and reliance on “utilizing ChatGPT to study” appears to be a straightforward, efficient act, yet we know this

is not true. AI conceals a heterogeneous network of training data, constraints, institutional policies, ethical norms and interpretive judgment with the overall question of validity, whether AI provides useful knowledge or reprocessed partial facts. Non-human intermediaries, such as internet infrastructure and laptops, quietly transfer information within a network and become visible only when they fail. AI is triggered in response to time pressures reinforced by part-time employment and commuting, acting as an actant that discourages slower, more traditional exploratory learning, further consolidating this ordering as technology only grows.

There is an asymmetric power distributed in this network. Failure and Success are depicted by institutional actors' grading schemes, assessment structures, and academic policies that ultimately affect future employment. As for procedural power, Generative AI runs as a powerhouse through its speed, availability and capacity to translate complex engineering content into accessible forms. My resistance lies with both validity and dependency, where reduced depth of learning is absorbed and obscured by punctualization. The ordering persists because it aligns actor interests: efficiency for the student, measurable outputs for the institution, and sustained use for the AI system.

2. Increased Awareness of AI Constraints (Recent Change / Last 12 Months)

Generative AI took the academic scene by storm and continually pushes this actor-network to reconfiguration due to increased visibility and regulation. Instructor directives, revised academic integrity policies, and heightened discourse around AI misuse have modified AI from an invisible intermediary into a visible and contested mediator. This shift produced a new perspective, casting light on destabilizing the previous order, and the assumption that AI use was neutral or unproblematic.

This functioned as a de-punctualization of the network. Previously concealed actors' disciplinary committees and professional ethical standards became noteworthy. As a result, my relationship with Generative AI was translated. Rather than delegating solution generation, I now engage AI more selectively: requesting conceptual clarification, problem-solving strategies, or explanations of lecture material. This translation reasserts human agency while discouraging reliance on AI outputs.

New relations also emerged. Engineering ethics and professionalism gained influence, constraining acceptable uses of AI while simultaneously encouraging responsible engagement. Future employers became a more prominent actant, shaping my perception of what competencies must be developed rather than simulated. The network reorganized rather than collapsed. Power was redistributed back toward me as an active interpreter, while AI remained influential but increasingly constrained. This change highlights a core idea in ANT. What feels stable is actually fragile, and when relationships shift, hidden structures underneath become visible, and members of influence can change.

3. Generative AI (Non-Human Actor) with Power

Generative AI functions as a powerful non-human actor within my learning network due to its affordances and its alignment with dominant institutional actors. Through generalized symmetry, ANT allows this power to be analyzed without privileging human intent over non-human capacity. Generative AI provides rapidity, encourages efficiency, and supports task completion under time pressure. In coordination with deadlines, grades, and learning management systems, it contributes to the production of a recognizable "product" that is suitable for academic performance.

This power is sustained through normalization and convenience. The availability of AI reshapes expectations around learning speed and effort, discouraging prolonged struggle and shifting learning toward outcome-oriented behaviour. My initial resistance concerns about over-reliance or lack of conceptual depth were gradually shifted because AI doesn't replace learning outright. Rather, it mediates learning as a coordination between human judgment and machine-generated output.

Crucially, this power is relational rather than deterministic. Generative AI's influence depends on time pressure, institutional incentives, and my willingness to delegate cognitive labour. Academic integrity policies and professional ethics operate as counter-forces that limit how far this power can extend. ANT reveals that AI's dominance is not inevitable. It is produced, maintained, and potentially resisted through relationships. If assessment structures or workload distributions were reorganized, the network and its power dynamics could be reorganized differently.